

CLINICAL INVESTIGATION

Eyes

RADIATION COMPLICATIONS AND TUMOR CONTROL AFTER ^{125}I PLAQUE BRACHYTHERAPY FOR OCULAR MELANOMA

ASHLEY W. JENSEN, B.A.,* IVY A. PETERSEN, M.D.,[†] ROBERT W. KLINE, PH.D.,[†]
SCOTT L. STAFFORD, M.D.,[†] PAULA J. SCHOMBERG, M.D.,[†] AND DENNIS M. ROBERTSON, M.D.[‡]

*Department of Radiation Oncology, Mayo Medical School, Mayo Clinic College of Medicine, Rochester, MN; and [†]Division of Radiation Oncology and [‡]Department of Ophthalmology, Mayo Clinic, Rochester, MN

Purpose: To determine the outcome of ^{125}I plaque brachytherapy at our institution and identify the risk factors associated with the development of radiation complications, tumor recurrence, and metastasis.

Patients and Methods: From 1986 to 2000, 156 patients underwent ^{125}I episcleral plaque (COMS design) application for the treatment of ocular melanoma. Chart analysis of follow-up ophthalmologic appointments assessed the incidence of ocular side effects after therapy. Statistical analysis assessed outcomes and significant influencing factors.

Results: With a median follow-up of 6.2 years, the 5-year overall survival was 83%. The 5-year disease-specific survival was 91%. Initial local control at 5 years was 92%, with 100% ultimate local control after secondary therapy that included 9 enucleations. The risk of metastasis was 10% at 5 years and 27% at 10 years. Vision stayed the same or improved in 25% of patients, and 44% of patients maintained visual acuity better than 20/200. Thirteen percent of patients experienced chronic pain or discomfort in the treated eye. Dose rates to the tumor apex greater than 90 to 100 cGy/h were associated with increased systemic control but worse radiation toxicity. **Conclusion:** Patients in our series experienced excellent local tumor control. Higher dose rates to the tumor apex were associated with reduced rates of distant metastases but worse ocular function. © 2005 Elsevier Inc.

Ocular melanoma, Plaque brachytherapy, ^{125}I , Outcome.

INTRODUCTION

Choroidal melanoma is a rare but deadly ocular malignancy. Untreated ocular melanoma has a high rate of systemic metastasis. As no cure exists for systemic spread, death typically ensues within months. The focus of treatment is to halt progression of the tumor and limit it to the eye. The standard therapy for more than 100 years was surgical enucleation, which drastically improved tumor control and overall survival. Modern techniques of radiation therapy for ocular melanoma have resulted in survival rates at least as good as with enucleation (1). Episcleral plaque brachytherapy by use of isotopes such as ^{60}Co , ^{106}Ru , and ^{125}I have achieved local control rates in the range of 88% to 98% at 5 years (2–5). Trials with charged particles such as helium ions and proton beams have been even more promising; reported local control rates were as high as 99% at 5-years (5, 6).

Plaque brachytherapy with ^{125}I is currently the most common treatment modality for ocular melanoma in the

United States. Despite the favorable dose distribution of ^{125}I , this therapy carries a high rate of delayed radiation damage to the retina, optic nerve, and other ipsilateral eye structures. Patients often experience variable amounts of decline or loss of visual acuity from cataract formation and other ocular morbidities over the subsequent 1 to 3 years. However, radiation therapy has potential advantages over enucleation, such as delay or reduction of vision loss and a more favorable cosmetic effect.

An ongoing controversy existed for years as to whether radiation therapy was as effective as enucleation at halting local progression and decreasing mortality. This question was definitively answered by the Collaborative Ocular Melanoma Study (COMS). This series of multicenter, randomized controlled trials enrolled and treated more than 1,300 patients with medium-sized choroidal melanoma and reported no significant 5-year survival advantage of enucleation versus ^{125}I irradiation (7). However, although the COMS group answered the question of survival versus treatment modality, it did not address certain strategies of

Reprint requests to: Ivy A. Petersen, M.D., Division of Radiation Oncology, Mayo Clinic, 200 First Street SW, Rochester, MN 55905. Tel: (507) 284-2669; Fax: (507) 284-0079. E-mail: petersen.ivy@mayo.edu

Presented at the 90th Scientific Assembly and Annual Meeting of The Radiological Society of North America, Chicago, IL,

November 29, 2004.

Acknowledgment—The authors thank the statisticians at the Mayo Clinic Center for Patient-Oriented Research for their invaluable statistical assistance.

Received Sep 27, 2004, and in revised form Jan 4, 2005. Accepted for publication Jan 11, 2005.